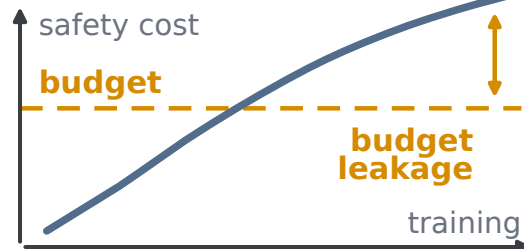


Penalty-based safe RL

$$\max_{\pi} J_r(\pi) - \lambda^\top \mathbf{J}_c(\pi)$$



Typical failure mode

One penalty vector prices all channels;
a rare channel gets traded away.

SafeTransport

one capacitated γ -flow bridge

1. Encode

2. Compose

3. Solve once

c_1 **Bridge impact**
hard $S^{(1)}$

c_2 **Liquid spill**
soft $S^{(2)}$

c_3 **Contact force**
hard $S^{(3)}$

c_K **Fire / tipping**
hard $S^{(K)}$

Hard minimum

$$S_{ss'}^{(k)} = \frac{b_k \omega_{ss'}^{(k)}}{\bar{c}_k(s)}$$

$$S = \min_k S^{(k)}$$

certificate

smooth Compose_{T_c}

$S^{(1)}$

$S^{(2)}$

$S^{(K)}$

**capacitated
 γ -flow Sinkhorn**

$$F_\varepsilon^* = \arg \min_{F \in \mathcal{F}_S}$$

$$\langle D, F \rangle - \varepsilon H(F)$$

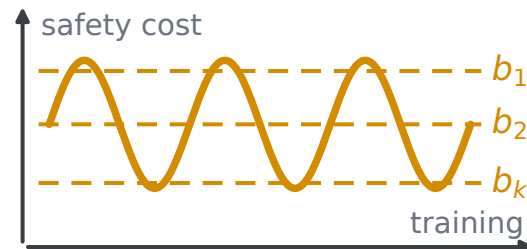
feasible target flow



$$F_\varepsilon^* \leq S$$

Constraint-based safe RL

$$\max_{\pi} J_r(\pi) \quad \text{s.t.} \quad J_{c_k}(\pi) \leq b_k \quad \forall k$$



Typical failure mode

Coupled dual updates oscillate
near several active budgets.

Conceptual positioning: penalties $\rightarrow$ one SafeTransport capacity solve $\rightarrow$ projections

PPO / SAC

PPO-Lag / FOCOPS

SafeTransport

one inner capacity projection

CPO

CUP / hard projection

unconstrained or penalty

external safety controller

local constrained step

constraint-dominant

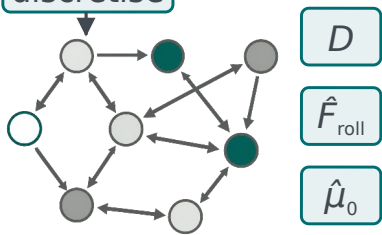
FLOW SPACE

budgets b_k

$b_k, k = 1 \dots K$

2 Flow graph

discretise



$\omega^{(k)}, \bar{c}_k$

3 Capacity compile

$S^{(1)}$

$S^{(2)}$

$\vdots$

$S^{(K)}$

$$S^{(k)} = b_k \omega^{(k)} / \bar{c}_k$$

$S^{(1 \dots K)}$

4 Compose

hard min
(certificate)

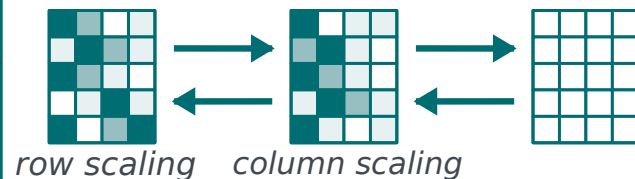
softmin T_c

S
(capacity
ceiling)

5 γ -Flow Sinkhorn

$$\min \langle D, F \rangle - \varepsilon H(F)$$

ε



$$0 \leq F \leq S \quad F \mathbf{1} - \gamma F^T \mathbf{1} = (1 - \gamma) \mu_0$$

F^*

transitions, rewards,
 K costs

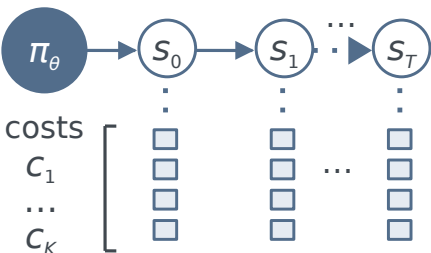
$D, \hat{\mu}_0$

gates

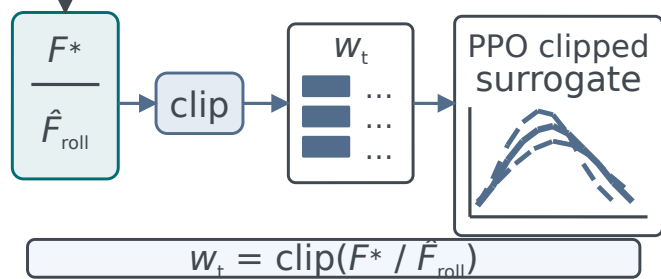
F^*

POLICY SPACE

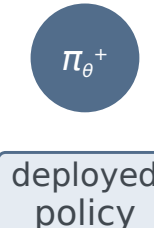
1 Policy rollout



6 Flow-weighted PPO

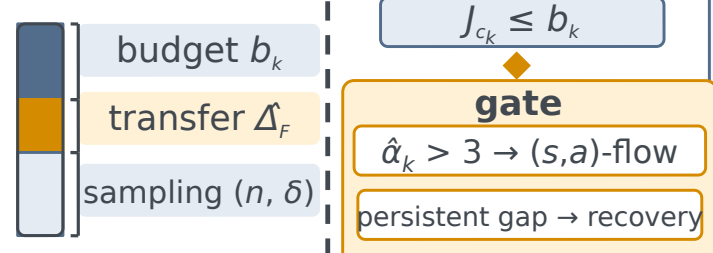


$$w_t = \text{clip}(F^* / \hat{F}_{\text{roll}})$$



π_{θ^+}
 $\hat{F}_{\theta^+}, J_{c_k}$

7 Deployed-policy audit



$$J_{c_k} \leq b_k$$

gate

$\hat{\alpha}_k > 3 \rightarrow (s, a)\text{-flow}$

persistent gap $\rightarrow$ recovery

next rollout